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Mechanically Recycling Metallicized Fabrics

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Electronic textiles (e-textiles) are a problematic technology when designing for sustainability as they combine two of the highly polluting industries - textiles and electronics. While some sustainability strategies have been proposed to combat their end-of-life issues, what happens after components are separated and whether they can still function in new contexts has not been sufficiently addressed. Metallicized fabrics are commonly used in the e-textiles space for prototyping to provide conductivity for an array of applications, thus this study is focused on the post-disassembly potential and limitations of these materials. We apply recycling techniques from conventional mechanical textile recycling to produce needle-punched nonwoven fabrics. Through the characterization of the new nonwovens and the exploration of piezoresistive properties we advocate for a more nuanced understanding of post-disassembly reuse and for expanding sustainable design for e-textiles.

Keywords: e-textiles; nonwoven; sustainability; pressure sensors

1. Introduction

E-textiles represent fabrics with diverse functions enabled by technological components, a portion of which are embedded within the textile structure itself (Beeby et al., 2025). In functions requiring electrical conductivity, materials such as metal wires, metal-coated yarns, carbonized fibers, and carbon black are used (Leis, 2020). E-textiles have emerged as a significant sector because they integrate computational capabilities into textiles, a material deeply embedded in everyday life. Their inherent softness, flexibility, and wearability support novel interaction modalities that move beyond screen-based paradigms, enabling embodied, tangible, and multisensory experiences. By embedding sensing, actuation, and communication directly in fabrics, e-textiles facilitate seamless interactions

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that are both functional and expressive (Aigner et al., 2024; Luo et al., 2021; Rallabandi et al., 2024; Wang et al., 2025; Zhou et al., 2023). Yet alongside this growing popularity, e-textiles pose acute sustainability challenges. Their very strength, the seamless integration of conductive elements with textile substrates, also makes them environmentally problematic.

However, the hybrid nature of conductive textiles complicates both reuse and recycling, since they integrate diverse material types such as conductive polymers, metallic coatings, filaments, and nanomaterials (Dulal et al., 2025; Eppinger et al., 2023; Köhler et al., 2011; Leis, 2020). Researchers have begun to explore strategies to address these sustainability challenges, ranging from the development of bio-based conductive materials to novel recycling methods (Dulal et al., 2025; Guridi et al., 2024; Zhu et al., 2024). Bio-based or biodegradable conductive materials can decompose safely at end-of-life reducing environmental impact, but the materials enabling this often don't offer the same functionality and longevity as their metallicized counterparts (Abdi et al., 2024; Dulal et al., 2022; Jaiswal et al., 2023; Koelle et al., 2022; Van Der Velden et al., 2015; Zhu et al., 2024). Among these, design for disassembly (DfD) has become a go-to sustainability approach, promoting the idea that e-textiles can be dismantled into reusable or recyclable components (Beeby et al., 2025; Song et al., 2024; Wu & Devendorf, 2020). According to Forst, DfD also support the upgrading, repair, or adaptation of existing materials and products, thereby extending their useful lifetime and delaying the need for resource recovery (Forst, 2018).

Despite increasing recognition of the need for reuse and recycling in the e-textile sector, methods for reclaiming conductive textiles remain rare (Tadesse et al., 2025). While experimental work in chemical recycling exists, there have been no reported attempts at mechanical recycling of conductive textiles, in part due to the difficulty of separating integrated textile and electronic components (Dulal et al., 2024; Leis, 2020).

In this paper, we present a method for mechanically recycling conductive textiles and reforming them into conductive nonwovens for interactive applications (see Fig. 1). Through shredding and carding, we reclaim conductive fibers from discarded conductive textiles, which are then formed into nonwovens through needle-punching. This approach aligns with Forst's perspective on material longevity, where materials may transition into new states and applications before their recovery into mono-material streams through e.g. chemical recycling processes. In this context, reprocessing the metallicized fabrics into conductive nonwovens can be regarded as an intermediate strategy.

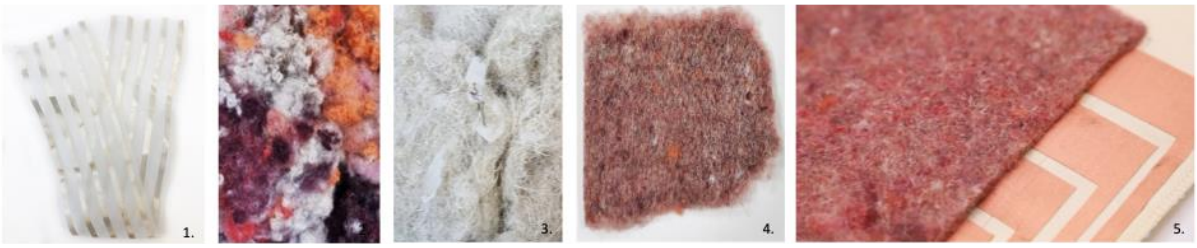


Figure 1. 1. Conductive Textile (Ag-Fil) before shredding, 2. Shredded Wool, 3. Shredded Conductive Textile, 4. Nonwoven Blend (Ag-Fil, Wool), 5. Downcycled Nonwoven Sensor (Ag-Fil-Wool)

2. Background

2.1. End-of-Life Barriers in E-Textiles

Both the electronics and textile industries are among the largest contributors to global pollution, with textiles alone responsible for vast water consumption, chemical use, and waste (Duhoux et al., 2022; Mikucioniene et al., 2024; Wojnowska-Baryła et al., 2024), and electronics contributing heavily to hazardous e-waste streams (Kaya, 2016; Rene et al., 2021; Shittu et al., 2021; Tansel, 2017; UN, 2017). The Ellen MacArthur Foundation estimates 134 million tons globally of discarded textiles by 2030 (Ellen Mac Arthur Foundation, 2021). Similarly, the consumption of electrical and electronic equipment (EEE) is rising quickly, while the average lifespan of the products has significantly decreased (Kaya, 2016). Projections from the Global E-waste Monitor 2024 by the United Nations Institute for Training and Research estimate that e-waste will reach 82 million metric tons globally by 2030 (Baldé et al., 2024). The e-textile market is projected to grow to USD 1.4 billion by 2030 (Dulal et al., 2025) and it has been found that 28.6% of textile and electronics recycling companies are already reporting experience with processing e-textiles (Eppinger et al., 2023). The hybrid material composition poses additional barriers to recycling, as it complicates separation and frequently causes degradation of electronic or conductive functionality during conventional processes. Understanding the lifecycle and recyclability of e-textiles is thus critical to developing sustainable design practices.

Currently, the textile and electronics sectors operate largely independently across their entire value chains, from production to end-of-life processing (Köhler et al., 2011), thus e-textiles present an even greater challenge, as they combine characteristics from both sectors. Life-cycle assessment studies have begun to quantify these challenges (Dulal et al., 2025; Hamidi & Baljko, 2014), highlighting the need for scalable, practical solutions rather than proof-of-concept prototypes alone. Despite progress, many of these strategies remain limited in scope or have yet to demonstrate viability in real-world interactive products (Schischke et al., 2020).

2.2. Challenges in E-Textile Recycling

Design for recycling (DfR) is a well-established concept in the conventional textile industry that incorporates recyclability considerations at the design stage. However, this approach remains largely underexplored in the e-textiles domain, where the recycling of composite materials continues to require significant technological advancements and research (Eppinger et al., 2023).

Currently there is no established e-textile recycling method, so we look to the textile and electronics industries to see how the components are individually processed.

Due to their distinct mechanical properties, e-textiles are generally incompatible with conventional Waste Electrical and Electronic Equipment (WEEE) shredding machines, which are not designed to handle soft or flexible textile materials (Leis, 2020). Another concern is improper disposal through household waste streams, where incineration can lead to the release of toxic substances (Eppinger et al., 2023). It even remains uncertain to what extent current textile recycling machinery can effectively process metallicized yarns, or whether their presence may lead to increased mechanical wear and operational degradation. In summary, recycling and disposal of e-textile waste remain challenging, as current regulations, material classifications, and processing technologies are not yet equipped to handle this emerging waste stream (Eppinger et al., 2023; Köhler et al., 2011).

Mechanical and chemical recycling are the most common methods implemented in the textile industry (Baloyi et al., 2024; Damayanti et al., 2021; Juanga-Labayen et al., 2022). Chemical recycling involves the depolymerization of polymers, a process that enables the separation of blended materials. A number of studies have explored chemical recycling of conductive textiles (Dulal et al., 2024; Leis, 2020), but there is a significant gap regarding the mechanical recycling of these textiles. Mechanical recycling involves a process of separating, cutting, shredding, and carding textiles to obtain fibers to be re-used to make new yarns or nonwoven textiles. This is an established waste processing stream, but it typically yields lower-quality outputs due to fiber shortening and damage during processing. Consequently, it is often applied in the downcycling of fabrics into products such as packaging blankets or insulation materials (Baloyi et al., 2024; Becker et al., 2024; Zandberga et al., 2023). Despite the growing awareness of sustainability concerns in the expanding e-textile sector, practical recycling efforts remain rare. While some experimental studies in chemical recycling exist, no documented attempts at mechanical recycling have been reported, likely due to the challenge of separating tightly integrated electronic and textile components.

In this work we have chosen to explore how mechanical recycling can be applied to the recycling of metallicized woven fabrics designed for use in e-textiles.

3. Recycling Metallicized Textiles into Non-Woven Piezoresistive Sensors

Here we explore how established mechanical recycling techniques can be applied to metallicized textiles and reintegrated to produce new functional textiles. We do this through shredding and carding, as shown in Fig. 2. Needle-punching was carried out to fabricate the nonwovens, as this has been previously demonstrated (Bae & West, 2021; Devendorf et al., 2022), so can be successfully replicated using industrial machines.

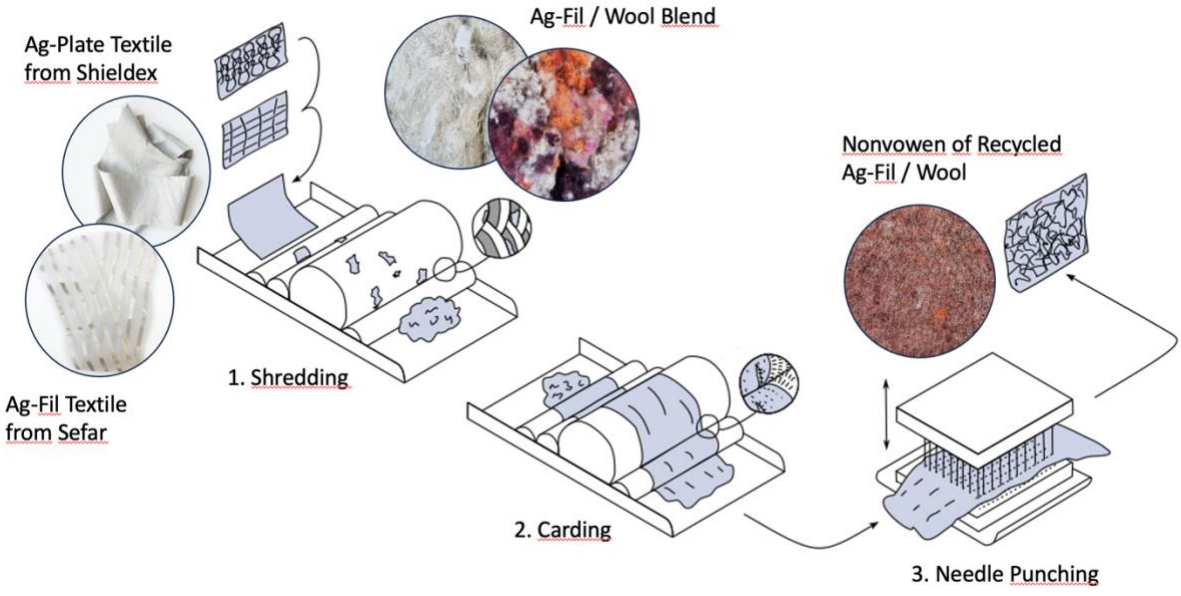


Figure 2. Illustration of the recycling process: 1. the shredding machine takes source fabrics and outputs textile pieces and neps of fibers; 2. the carding process combs the shredded material into a loose collections of aligned fibers; 3. the needle-punching machine agitates the fibers with barbed needles causing the fibers to interlock into a nonwoven textile.

3.1. Materials

We selected three conductive textiles to use as the source materials. All three textiles are woven and are made conductive by the addition of metal either by incorporating metal filaments into the weft or by plating non-conductive threads with metal. Additionally, we investigated the blending of shredded conductive fibers with recycled wool fibers to evaluate their impact on conductivity and material performance. All source textiles are shown in Table 1.

Two of the textiles manufactured by Sefar were materials from previous research prototypes donated to this work (Cheng et al., 2016; Zhou, Altamirano, et al., 2017; Zhou, Cheng, et al., 2017). The textile from Shieldex was purchased for this study. All sourced textiles are considered pre-consumer waste as they were not previously integrated in e-textile devices.

Table 1. Description of textiles used as source materials to be shredded.

Abbreviation	**Materials**	**Supplier**
Ag-Plate	Bremen - Plated Silver Polyamide	Shieldex
Ag-Fil	Silver Filaments with Polyester	Sefar
Zn-Fil	Zinc Filaments with Polyamide	Sefar
Ag-Fil-Wool	Silver Filaments with Polyester and Wool	Sefar

3.2. Mechanical Recycling of Conductive Fabrics

The recycling process consists of three sequential stages - shredding, carding, and needle-punching - each contributing to the transformation of conductive woven fabrics into recycled nonwovens. Fig. 3 shows the machines we used.

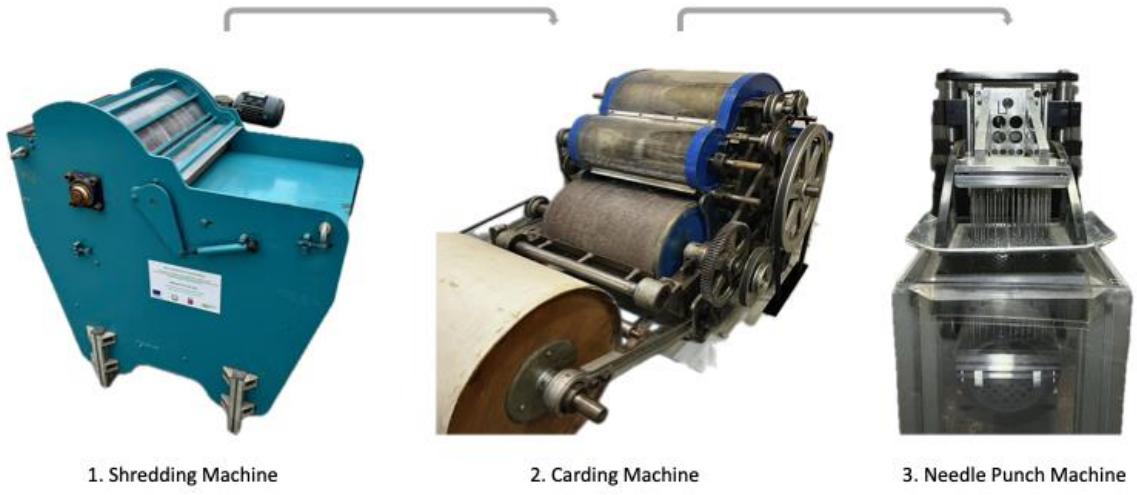


Figure 3. Machines used in the mechanical recycling process: 1. shredding machine; 2. carding machine; 3. needle punch machine.

3.2.1 Shredding

The recycling process begins with shredding in which woven fabrics are broken down into shorter staple fibers. These staple fibers then form the basis for subsequent carding and blending steps. A

shredding machine, containing a conveyor belt, pressure roller, and a rotating cylinder with small blades, gradually reduces the textile into staple fibers. The process was conducted at 230 V and 40 Hz, corresponding to 1100 rpm.

The shredding results in a loss of material as some of the fibers become overly shortened and are drop out during the process. Table 2 summarizes the weight loss for each of the textiles.

Table 2. Weight loss from shredding.

Sample	Starting weight (g)	Weight after shredding (g)	Weight loss (%)
Ag-Plate	36.7	29.0	21
Ag-Fil	166.2	103.9	37.5
Zn-Fil	70.6	52.5	25.6

3.2.2 Carding

Carding aligns the staple fibers into a continuous web with greater homogeneity. The shredded fibers were transferred onto a 30 cm wide conveyor belt, passing over a cylinder lined with fine metal teeth. Surrounding rollers guided the fibers until they were deposited as a thin, continuous web on the collection cylinder. The staple fibers (Ag-Plate, Ag-Fil and Zn-Fil) were each carded once, and some Ag-Plate staple fibers were carded twice to enhance uniformity. Some of the Ag-Fil fibers were mixed with recycled wool fibers. The final output was a homogeneous web with a thickness of 1 cm.

3.2.3 Needle-Punching

The final step is needle-punching, in which the carded web is consolidated into a nonwoven textile. Specialized needles (15X32X40X3 1/2 R222 G 530P7, Groz-Becker) mechanically entangle the fibers by drawing surface filaments downward into underlying layers. This action increases the cohesion of the web, producing robust nonwoven fabrics.

The 110 x 30 cm carded web from Ag-Fil, Zn-Fil and Ag-Fil-Wool was folded once and needle-punched. To increase the thickness to 5mm, the Ag-Plate was folded in three successive passes. After each cycle, the web was folded further, reducing its dimensions while increasing density and entanglement.

Out of the initial 100 x 110 cm of the recycled Ag-Plate, two 15 x 15 cm samples of 5 mm thickness were produced. During processing, significant needle breakages occurred, attributed to the tensile strength of conductive fibers and the stress of penetrating multiple folded layers. This mechanical wear highlights the challenges of processing conductive fabrics using traditional nonwoven equipment. The produced nonwovens are shown in Fig. 4.



(a) Ag-Fil *(b) Ag-Fil-Wool* *(c) Ag-Plate* *(d) Zn-Fil*

Figure 4. Needle punched recycled nonwoven e-textiles.

3.3. Electro-Mechanical Characterization

The materials were initially designed for use in piezoresistive sensor technology, and their characterization was conducted in line with this intended application. We assessed the sheet resistivity, pressure-sensing range, dynamic sensing behavior, and repeatability of the four samples. We present both the test methodology and the results below and discuss the implications of these results in section 4.

Measuring electrical resistance in nonwoven metallicized textiles presented unique challenges due to their structural and compositional irregularity. Unlike conductive threads or coated conductive fabrics, these nonwoven samples contain conductive elements dispersed unevenly throughout a fibrous matrix, as shown in Fig. 4. This randomness, combined with inconsistencies introduced by the needle-punching process, such as variable fiber orientation, thickness, and conductive fiber clustering, made conventional conductivity measurement techniques unreliable or inapplicable.

3.3.1 Sheet Resistivity

To overcome the dual challenges of low-resistance measurement and material inhomogeneity, we employed an adapted four-point probe method. This approach offers high accuracy for resistances below 1 Ω and has been successfully applied in prior studies on needle-punched nonwoven metallicized textiles composed of loose fibers (Lu et al., 2016; Pakdel et al., 2021). The protocol was tailored to accommodate the irregular geometry and porosity of the recycled nonwoven samples.

Measurements were taken using a Keithley 6500 digital multimeter with the Kickstart software (Ver 2.11.3), connected to a linear in-line four-point probe (probe spacing: 3 mm). A constant 1 N contact force was applied to ensure repeatable pressure across all trials. Resistance values were averaged over 60 seconds to minimize noise, with measurements taken at five locations per sample under ambient conditions (approximately 25◦C).

To correct for finite sample size, we applied a geometry-specific correction factor (C = 0.5920) to the classical four-point probe model (Smits, 1991; Valdes, 1954):

$$p = C\frac{V}{\mathrm{I}}$$

Where ρ is surface resistivity (in $\Omega{\cdot}m$), V is the measured voltage, and I is applied current. This adapted protocol provided reproducible surface conductivity measurements for non-uniform, recycled nonwoven conductive textiles.

The resistivity and conductivity values of the nonwoven samples are summarized in Table 3.

Table 3. Comparison of source textiles and the recycled nonwovens including the percentage loss of conductivity through the process.

Sample	**Source Material**		**Recycled Nonwoven**		
	Resistivity (Ω/m)	Conductivity (S/m)	Resistivity (Ω/m)	Conductivity (S/m)	Cond. Retained (%)
Ag-Plate	0.0000953	10500	0.0188	53.3	0.5
Ag-Fil	0.00000698	143266	0.0092	108	0.07
Zn-Fil	0.00000851	118000	0.0160	62.5	0.05

3.3.2 Pressure Range

To determine the pressure-sensing range of the recycled nonwovens, we applied a controlled load up to 400 N at a rate of 1 N/s while simultaneously measuring electrical resistance. Loading was performed on an Instron 5866 screw-driven uniaxial testing machine, with resistance recorded using a Keithley 6500 1/2 digital multimeter and Kickstart software.

The results of the fully tested range can be seen in Fig. 5. The roughly linearly responsive range for each sample was: Ag-Plate 0-60 N; Ag-Fil 10-150 N; Ag-Fil-Wool 0-30 N; and Zn-Fil 10-120 N.

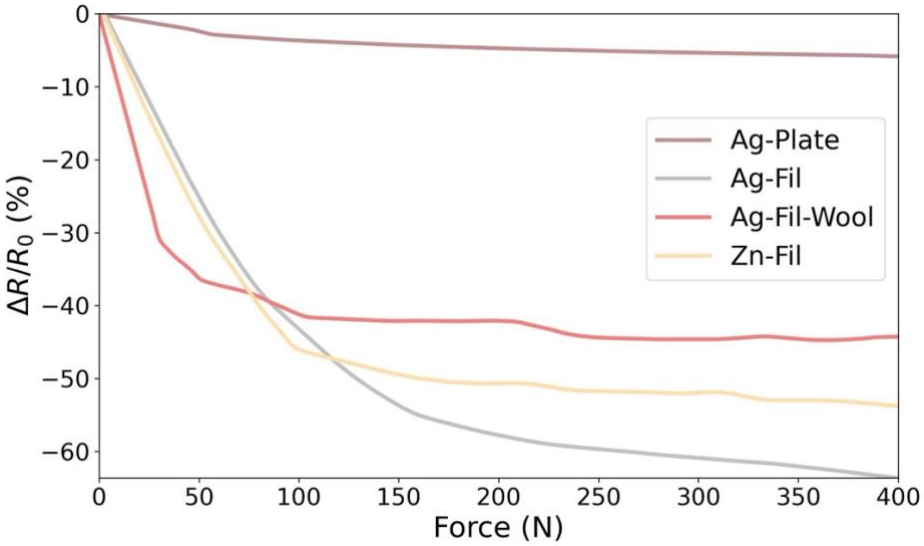


Figure 5. Monotonic compressive loading to 400N for conductive recycled nonwovens.

3.3.3 Dynamic Response Behavior

We characterized the dynamic response by applying an instantaneous 200 N force, holding for 5 s, and then releasing. This was done using the same set-up described in 3.3.2. A force of 200 N was selected based on the pressure range testing, as the linear portion. All of the samples were below that force level. Fig. 6 shows the results of the pressing, holding, and releasing.

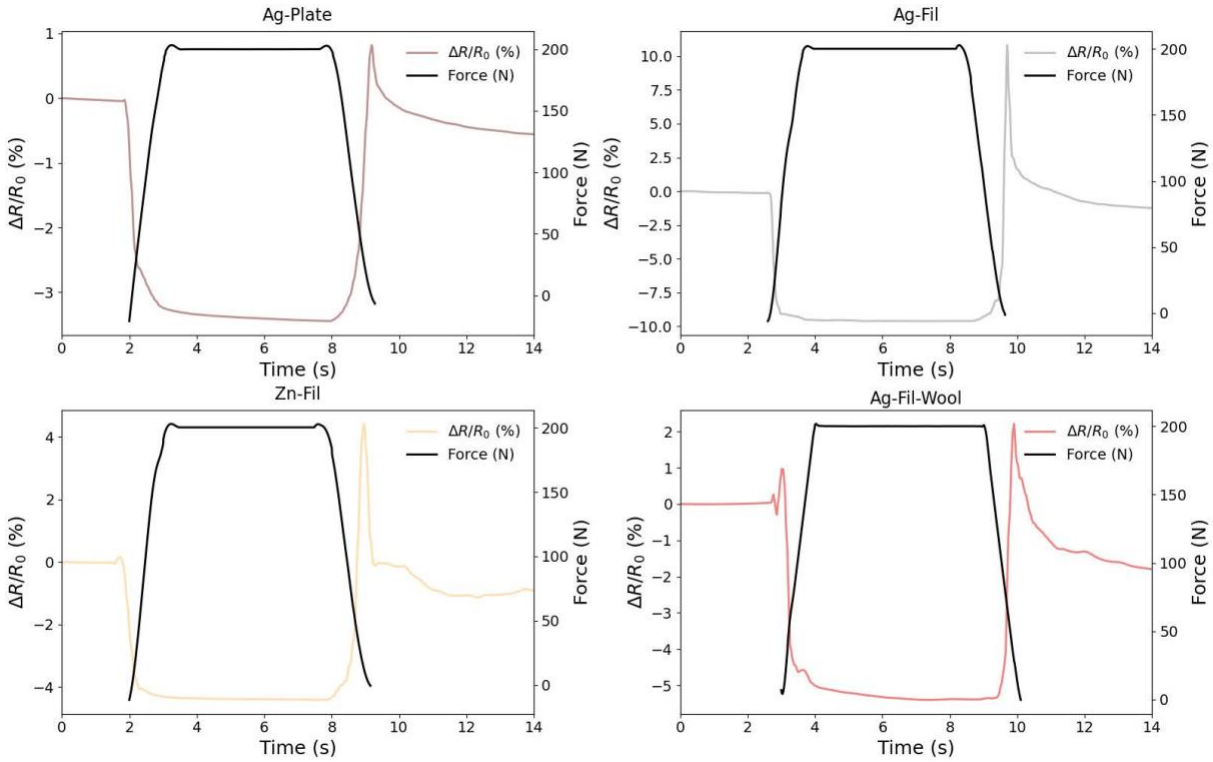


Figure 6. Dynamic testing showing response/recovery for nonwovens at 200N.

To assess durability and repeatability, cyclic compression tests up to 200N were conducted for 1000 loading-unloading cycles at 100 N/s - a little over 1.5 hours of continuous testing for each sample. Results from two of the samples can be seen in Fig. 7. All samples saw the greatest variation in the first 100 cycles (conditioning stage), before a more consistent, though reduced, resistance range was produced. The plated samples had a more consistent response over time than the filament samples. All samples continued to exhibit a piezoresistive response to applied pressure at the end of the testing.

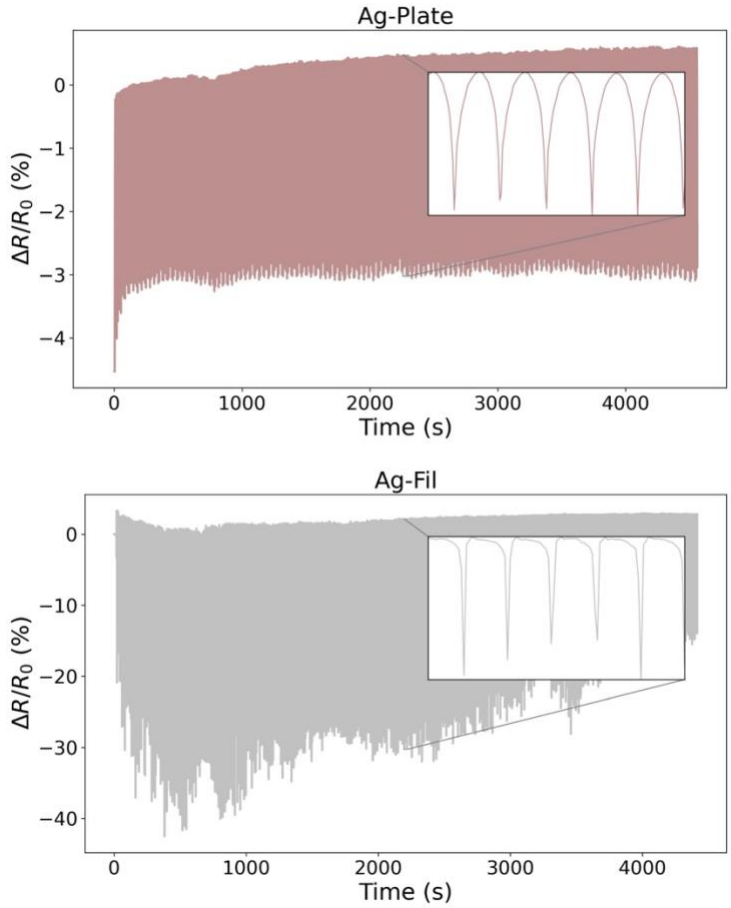


Figure 7. Repeatability and durability of nonwoven sensors over 1000 compression cycles.

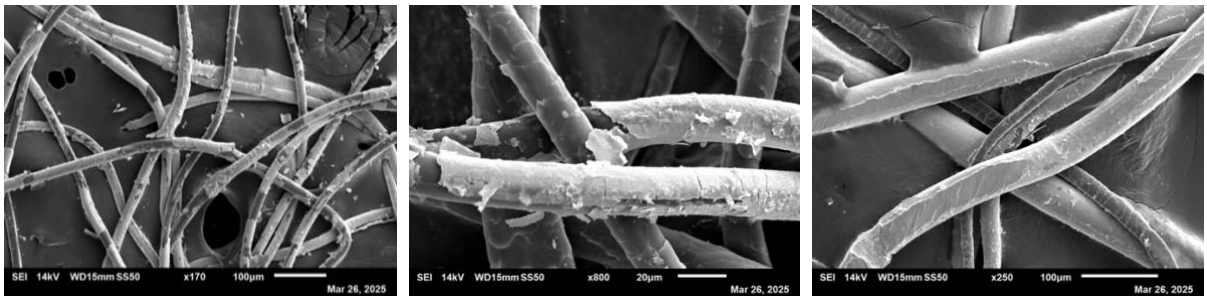
4 Discussion

Our findings show that mechanical recycling of conductive textiles substantially reduces conductivity across all tested samples, presenting both challenges and opportunities for sustainable e-textile design. This section discusses the implications of these results for interactive systems, highlighting how they intersect with questions of material degradation and design paths for sustainability.

4.1 Material Degradation

The most significant effect of mechanical recycling was the dramatic loss of conductivity across all textiles (Table 3). The Ag-Plate nonwoven retained only 0.5% of its original conductivity, while metallic filament fabrics (Ag-Fil, Zn-Fil) retained only 0.05%. This difference can be traced to the conductive-to-nonconductive ratios in their construction. In plated textiles, the entire fiber surface is coated with silver, providing initial continuous pathways, whereas filament-based fabrics interweave discrete conductive and nonconductive yarns. Consequently, the plated material had more conductive volume to lose, and its residual conductivity, though severely diminished, remained marginally higher. The mechanisms of degradation differ between conductive textile types. In filament fabrics, needle punching and shredding dented and fractured metallic filaments, leaving them intact but permanently deformed. SEM imaging confirmed that filaments remained metallic, albeit scarred. In contrast, plated textiles exhibited delamination, with the silver layer flaking away from the polyamide core (Fig. 8). During recycling, this coating detached almost as a dust, meaning that significant amounts of conductive material were lost to abrasion rather than redistributed into the nonwoven matrix. This

suggests that coated conductive textiles inherently degrade more under mechanical stress than those constructed from integrated metallic filaments.



(a) Recycled Ag-Plate. *(b) Recycled Ag-Plate at a greater* *(c) Recycled Zn-Fil.*

Figure 8. Scanning electron microscope micrographs showing damage to conductive elements after recycling.

While our experiments focused on piezoresistive sensing, other sensing modalities (e.g., capacitive or inductive approaches) might leverage residual conductivity more effectively, particularly where continuous conductive paths are less critical. Nonetheless, the results point to the broader challenge: conductivity is not indefinitely recoverable after disassembly.

4.2 Production Challenges

To produce a nonwoven textile of the desired thickness for the pressure-sensing demonstrators using the Ag-Plate sample, the material was folded and needle-punched after each folding step over three iterations. This resulted in a material with a high density of fibers, causing needles to break. This was not a prominent issue when needle punching only a single iteration, but this limits the nonwoven thickness that can be achieved.

To overcome this challenge, needle-punching equipment should be adapted to accommodate higher density materials containing metallicized fibers. While in our study we used a multi-purpose needle punch machine, there are other machines on the market that are specialized for high performance technical textiles applied in the automotive or construction industries. Given that these machines are already specialized for handling robust fibers such as glass or carbon, they may be more suitable for processing metallicized fibers.

4.3 Limitations of Nonwovens

The nonwovens produced from recycled fibers exhibited several material constraints relevant to being incorporated into interactive devices. First, their fragility and “dustiness” make them unsuitable for direct skin contact. The friable nature of fibers, especially where coatings had shed, created a rough hand-feel and visible particulate shedding, suggesting the need for encapsulation layers in any wearable application. To further overcome the shedding of fibers and counter the dusty feel of the nonwovens, designers could further explore the mixing of recycled conductive fibers with conventional softer fibers, such as recycled wool or synthetics. Our initial test suggested that physically separating the conductive fibers by incorporating nonconductive wool fibers enhanced the piezoresistive properties of the nonwovens. From a sensing perspective, the addition of wool fibers improved stability. The insulating fibers reduced the density of conductive contact points, reducing noise and making piezoresistive signals easier to process.

Mechanical loading revealed clear differences between textile types. Metallic filament nonwovens (Ag-Fil, Zn-Fil) ceased showing significant resistance change above approximately 200 N, saturating

under heavy loads. In contrast, the Ag-Plated nonwoven responded nearly linearly to applied force but only achieved a 5% resistance change, insufficient for detection in our tested circuit. With amplification circuitry, however, such materials could potentially still be used for low-range, linear sensing.

Durability testing also highlighted divergent lifespans among the nonwovens. In cyclic testing (Fig. 7), Ag-Fil showed progressive performance degradation, as metallic filaments underwent permanent plastic deformation, reducing their dynamic range. Ag-Plated, by contrast, retained its limited dynamic range more consistently, likely because the plated polyamide cores deformed elastically rather than plastically.

4.4 Exploration of Piezoresistive Sensing Potential

The nonwovens exhibited hysteresis and recovery delays of around 1 s, with incomplete return to baseline resistance after unloading. This settling period reflects fiber realignment and intermittent contact dynamics in the nonwoven structure. Such lag constrains their use for fast, dynamic interactions but is acceptable for applications like posture monitoring, where noise suppression is more valuable than speed.

Synthesizing these results, it is observed that the filament-based nonwovens displayed superior dynamic performance. However, these same filament-based materials showed higher variability and faster degradation during cyclic testing. Thus, there is a trade-off between dynamic range and stability: plated materials perform consistently but weakly, while filament-based fabrics offer stronger signals but less reliability over time.

Noise in resistance signals, arising from intermittent fiber-fiber contacts, further complicates integration. The recycled nonwovens are not sensitive enough for small-scale physiological signal sensing (e.g., pulse detection), but remain suitable for broader, spatial sensing applications such as posture monitoring or impact detection in protective gear.

4.5 Takeaways for Sustainable Pathways in E-Textile Design

Our study has produced downcycled textiles, functional but limited materials that may only be suited for specific classes of interactive applications.

To better understand how to broaden sustainability pathways for metallicized fabrics and improve material longevity, further studies are required. Our results illustrate that blending recycled conductive fibers with conventional fibers (e.g., wool) can complement piezoresistive behavior in new nonwovens while enabling new hybrid material possibilities. Some of our recent studies are investigating blends that move closer towards mono-material solutions, for example by incorporating polyamide or polyester instead of wool, depending on the base material of the metallicized textiles. Such blends open a space for designing “second-life” interactive materials: textiles intentionally downcycled for less demanding, yet still meaningful applications. In a subsequent step, the nonwovens could be processed through chemical recycling to separate and recover their mono-material components. In this context, mechanical recycling can be understood as an intermediate step that extends the useful lifetime of the metallicized textiles before their material recovery.

5. Conclusion

We have introduced an approach for recycling metallicized conductive textiles through mechanical recycling and needle-punching to create new pressure-sensing textiles for interactive applications. Through the recycling of three different source textiles, we have demonstrated that conductive textiles used in e-textile devices are prone to performance degradation during mechanical recycling, limiting their direct reuse in high-performance applications. As sustainable approaches are becoming even more important to every aspect of design, this work outlines pathways for extending the functional lifetime and value of e-textile components before their eventual recovery into mono-material streams through chemical recycling.

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